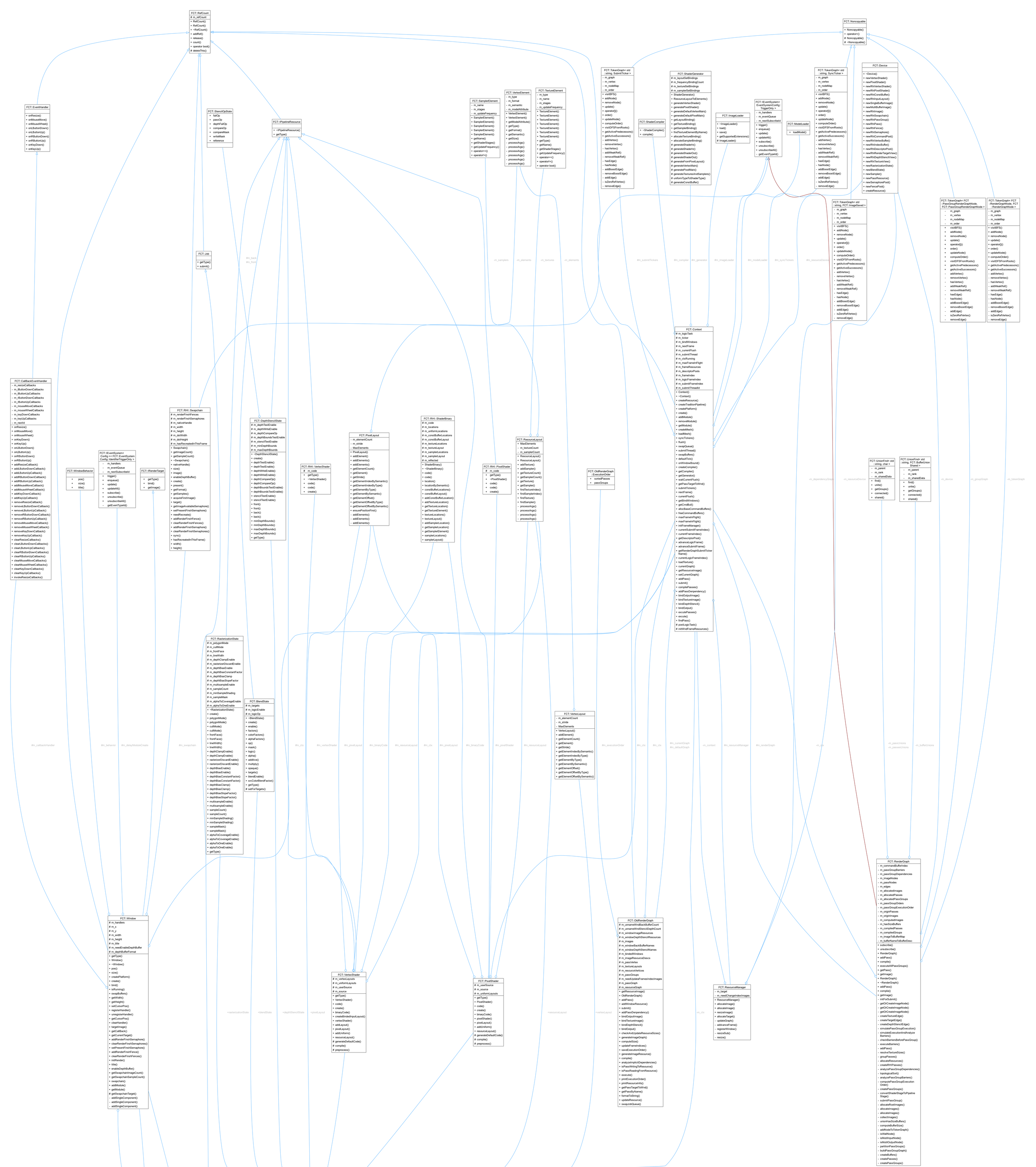
FCT::RefCount
m_refCount
+ RefCount()
+ RefCount()
+ ~RefCount()
+ addRef()
+ release()
+ count()
+ operator bool()
# deleteThis()
FCT::Noncopyable
+ Noncopyable()
+ operator=()
# Noncopyable()
# ~Noncopyable()
FCT::EventHandler
+ onResize()
+ onMouseMove()
+ onMouseWheel()
+ onLButtonDown()
+ onLButtonUp()
+ onRButtonDown()
+ onRButtonUp()
+ onKeyDown()
+ onKeyUp()
FCT::StencilOpState
+ failOp
+ passOp
+ depthFailOp
+ compareOp
+ compareMask
+ writeMask
+ reference
FCT::IPipelineResource
+ ~IPipelineResource()
+ getType()
FCT::SamplerElement
- m_name
- m_stages
- m_updateFrequency
+ SamplerElement()
+ SamplerElement()
+ SamplerElement()
+ SamplerElement()
+ getName()
+ getShaderStages()
+ getUpdateFrequency()
+ operator==()
+ operator!=()
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- m_format
- m_semantic
- m_modelAttribute
+ VertexElement()
+ VertexElement()
+ getModelAttribute()
+ getType()
+ getFormat()
+ getSemantic()
+ getSize()
- processArgs()
- processArgs()
- processArgs()
- processArgs()
- processArgs()
- processArgs()
FCT::TextureElement
- m_type
- m_name
- m_stages
- m_updateFrequency
+ TextureElement()
+ TextureElement()
+ TextureElement()
+ TextureElement()
+ TextureElement()
+ TextureElement()
+ TextureElement()
+ getType()
+ getName()
+ getShaderStages()
+ getUpdateFrequency()
+ operator==()
+ operator!=()
+ operator bool()
FCT::TokenGraph< std ::string, SubmitTicker >
- m_graph
- m_vertex
- m_nodeMap
- m_order
+ visitBFS()
+ addNode()
+ removeNode()
+ update()
+ operator[]()
+ order()
+ updateNode()
+ computeOrder()
+ visitDFSFromRoots()
+ getActivePredecessors()
+ getActiveSuccessors()
- addVertex()
- removeVertex()
- hasVertex()
- addWeakRef()
- removeWeakRef()
- hasEdge()
- hasNode()
- addBoostEdge()
- removeBoostEdge()
- addEdge()
- isZeroRefVertex()
- removeEdge()
FCT::ShaderCompiler
+ ~ShaderCompiler()
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FCT::ShaderGenerator
m_layoutSetBindings
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+ ShaderGenerator()
+ ResourceLayoutToElements()
+ generateVertexShader()
+ generatePixelShader()
+ generateDefaultVertexMain()
+ generateDefaultPixelMain()
+ getLayoutBinding()
+ getTextureBinding()
+ getSamplerBinding()
+ findTextureElementByName()
+ allocateTextureBinding()
+ allocateSamplerBinding()
# generateShaderIn()
# generateShaderIn()
# generateShaderOut()
# generateShaderOut()
# generateFromPixelLayout()
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# generatePixelMain()
# generateTexturesAndSamplers()
# uniformTypeToShaderType()
# generateConstBuffer()
FCT::ImageLoader
+ ~ImageLoader()
+ load()
+ save()
+ getSupportedExtensions()
# ImageLoader()
FCT::IEventSystem< EventSystemConfig: TriggerOnly >
- m_handlers
- m_eventQueue
- m_nextSubscribeId
+ trigger()
+ enqueue()
+ update()
+ updateAll()
+ subscribe()
+ unsubscribe()
+ unsubscribeAll()
- getEventTypeId()
FCT::ModelLoader
+ loadModel()
FCT::TokenGraph< std ::string, SyncTicker >
- m_graph
- m_vertex
- m_nodeMap
- m_order
+ visitBFS()
+ addNode()
+ removeNode()
+ update()
+ operator[]()
+ order()
+ updateNode()
+ computeOrder()
+ visitDFSFromRoots()
+ getActivePredecessors()
+ getActiveSuccessors()
- addVertex()
- removeVertex()
- hasVertex()
- addWeakRef()
- removeWeakRef()
- hasEdge()
- hasNode()
- addBoostEdge()
- removeBoostEdge()
- addEdge()
- isZeroRefVertex()
- removeEdge()
FCT::Device
+ ~Device()
+ newVertexShader()
+ newPixelShader()
+ newRhiVertexShader()
+ newRhiPixelShader()
+ newRhiConstBuffer()
+ newRhiInputLayout()
+ newSingleBufferImage()
+ newMutilBufferImage()
+ newRhiImage()
+ newRhiSwapchain()
+ newRhiPassGroup()
+ newRhiPass()
+ newRhiFence()
+ newRhiSemaphore()
+ newRhiCommandPool()
+ newRhiVertexBuffer()
+ newRhiIndexBuffer()
+ newRhiDescriptorPool()
+ newRhiRenderTargetView()
+ newRhiDepthStencilView()
+ newRhiTextureView()
+ newRasterizationState()
+ newBlendState()
+ newSampler()
+ newPassResource()
+ newSemaphorePool()
+ newFencePool()
+ createResource()
FCT::TokenGraph< FCT ::PassGroupRenderGraphNode, FCT::PassGroupRenderGraphNode >
- m_graph
- m_vertex
- m_nodeMap
- m_order
+ visitBFS()
+ addNode()
+ removeNode()
+ update()
+ operator[]()
+ order()
+ updateNode()
+ computeOrder()
+ visitDFSFromRoots()
+ getActivePredecessors()
+ getActiveSuccessors()
- addVertex()
- removeVertex()
- hasVertex()
- addWeakRef()
- removeWeakRef()
- hasEdge()
- hasNode()
- addBoostEdge()
- removeBoostEdge()
- addEdge()
- isZeroRefVertex()
- removeEdge()
FCT::TokenGraph< FCT ::RenderGraphNode, FCT ::RenderGraphNode >
- m_graph
- m_vertex
- m_nodeMap
- m_order
+ visitBFS()
+ addNode()
+ removeNode()
+ update()
+ operator[]()
+ order()
+ updateNode()
+ computeOrder()
+ visitDFSFromRoots()
+ getActivePredecessors()
+ getActiveSuccessors()
- addVertex()
- removeVertex()
- hasVertex()
- addWeakRef()
- removeWeakRef()
- hasEdge()
- hasNode()
- addBoostEdge()
- removeBoostEdge()
- addEdge()
- isZeroRefVertex()
- removeEdge()
FCT::TokenGraph< std ::string, FCT::ImageSaved >
- m_graph
- m_vertex
- m_nodeMap
- m_order
+ visitBFS()
+ addNode()
+ removeNode()
+ update()
+ operator[]()
+ order()
+ updateNode()
+ computeOrder()
+ visitDFSFromRoots()
+ getActivePredecessors()
+ getActiveSuccessors()
- addVertex()
- removeVertex()
- hasVertex()
- addWeakRef()
- removeWeakRef()
- hasEdge()
- hasNode()
- addBoostEdge()
- removeBoostEdge()
- addEdge()
- isZeroRefVertex()
- removeEdge()
FCT::Job
+ getType()
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FCT::Context
m_logicTask
m_ticker
m_bindWindows
m_nextFrame
m_currentFlush
m_submitThread
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m_maxFrameInFlight
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m_frameIndex
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+ Context()
+ ~Context()
+ createResource()
+ createTraditionPipeline()
+ createPlatform()
+ create()
+ addModule()
+ removeModule()
+ getModule()
+ createMesh()
+ loadMesh()
+ syncTickers()
+ flush()
+ swapQueue()
+ submitThread()
+ swapBuffers()
+ defaultTick()
+ onWindowBound()
+ createCompiler()
+ getCompiler()
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+ waitCurrentFlush()
+ getPassTargetToWnd()
+ submitTickers()
+ nextFrame()
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+ getBindWindows()
+ getCmdBuf()
+ allocBaseCommandBuffers()
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+ maxFrameInFlight()
+ maxFrameInFlight()
+ initFrameManager()
+ currentSubmitFrameIndex()
+ currentFrameIndex()
+ getDescriptorPool()
+ advanceLogicFrame()
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+ getRenderGraphSubmitTicker Name()
+ currentLogicFrameIndex()
+ loadTexture()
+ currentGraph()
+ getResourceImage()
+ setCurrentGraph()
+ addPass()
+ submit()
+ compilePasses()
+ addPassDenpendency()
+ bindOutputImage()
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+ excutePasses()
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# postLogicTask()
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FCT::CallBackEventHandler
- m_resizeCallbacks
- m_lButtonDownCallbacks
- m_lButtonUpCallbacks
- m_rButtonDownCallbacks
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- m_mouseMoveCallbacks
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- m_keyDownCallbacks
- m_keyUpCallbacks
- m_nextId
+ onResize()
+ onMouseMove()
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+ onLButtonUp()
+ onRButtonDown()
+ onRButtonUp()
+ addResizeCallback()
+ addLButtonDownCallback()
+ addLButtonUpCallback()
+ addRButtonDownCallback()
+ addRButtonUpCallback()
+ addMouseMoveCallback()
+ addMouseWheelCallback()
+ addKeyDownCallback()
+ addKeyUpCallback()
+ removeResizeCallback()
+ removeLButtonDownCallback()
+ removeLButtonUpCallback()
+ removeRButtonDownCallback()
+ removeRButtonUpCallback()
+ removeMouseMoveCallback()
+ removeMouseWheelCallback()
+ removeKeyDownCallback()
+ removeKeyUpCallback()
+ clearResizeCallbacks()
+ clearLButtonDownCallbacks()
+ clearLButtonUpCallbacks()
+ clearRButtonDownCallbacks()
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+ clearMouseMoveCallbacks()
+ clearMouseWheelCallbacks()
+ clearKeyDownCallbacks()
+ clearKeyUpCallbacks()
+ invokeResizeCallbacks()
FCT::WindowBehavior
+ pos()
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+ title()
FCT::IEventSystem< Config >< FCT::EventSystem Config::IdentifierTriggerOnly >
- m_handlers
- m_eventQueue
- m_nextSubscribeId
+ trigger()
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+ update()
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+ subscribe()
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FCT::IRenderTarget
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FCT::RHI::Swapchain
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+ Swapchain()
+ getImageCount()
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+ size()
+ image()
+ enableDepthBuffer()
+ create()
+ present()
+ getFormat()
+ getSamples()
+ acquireFirstImage()
+ target()
+ getImageAvailableSemaphore()
+ setPresentFinshSemaphore()
+ needRecreate()
+ addRenderFinshFence()
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+ sync()
+ hasRecreatedInThisFrame()
+ width()
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+ ~DepthStencilState()
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+ depthTestEnable()
+ depthWriteEnable()
+ depthWriteEnable()
+ depthCompareOp()
+ depthCompareOp()
+ depthBoundsTestEnable()
+ depthBoundsTestEnable()
+ stencilTestEnable()
+ stencilTestEnable()
+ front()
+ front()
+ back()
+ back()
+ minDepthBounds()
+ minDepthBounds()
+ maxDepthBounds()
+ maxDepthBounds()
+ getType()
FCT::RHI::VertexShader
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+ getType()
+ ~VertexShader()
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+ code()
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FCT::PixelLayout
- m_elementCount
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- MaxElements
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+ addElement()
+ addElements()
+ addElements()
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+ getStride()
+ getElementIndexBySemantic()
+ getElementIndexByType()
+ getElementBySemantic()
+ getElementBySemantic()
+ getElementOffset()
+ getElementOffsetByType()
+ getElementOffsetBySemantic()
+ ensurePositionFirst()
- addElements()
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FCT::RHI::ShaderBinary
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+ ShaderBinary()
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+ locationBySemantic()
+ constBufferLocations()
+ constBufferLayout()
+ addConstBufferLocation()
+ addTextureLocation()
+ getTextureLocation()
+ getTextureElement()
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+ addSamplerLocation()
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FCT::RHI::PixelShader
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+ ~PixelShader()
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FCT::ResourceLayout
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- m_textureCount
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+ ResourceLayout()
+ addTexture()
+ addSampler()
+ getTextureCount()
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+ getTexture()
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+ findTextureIndex()
+ findSamplerIndex()
+ findTexture()
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- processArgs()
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- processArgs()
FCT::OldRenderGraph ::ExecutionOrder
+ sortedPasses
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FCT::UnionFind< std ::string, char >
- m_parent
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FCT::UnionFind< std ::string, FCT::BufferUnion Shared >
- m_parent
- m_rank
- m_sharedData
+ find()
+ unite()
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FCT::RasterizationState
m_polygonMode
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m_frontFace
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m_depthClampEnable
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m_depthBiasEnable
m_depthBiasConstantFactor
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m_minSampleShading
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+ ~RasterizationState()
+ create()
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+ frontFace()
+ frontFace()
+ lineWidth()
+ lineWidth()
+ depthClampEnable()
+ depthClampEnable()
+ rasterizerDiscardEnable()
+ rasterizerDiscardEnable()
+ depthBiasEnable()
+ depthBiasEnable()
+ depthBiasConstantFactor()
+ depthBiasConstantFactor()
+ depthBiasClamp()
+ depthBiasClamp()
+ depthBiasSlopeFactor()
+ depthBiasSlopeFactor()
+ multisampleEnable()
+ multisampleEnable()
+ sampleCount()
+ sampleCount()
+ minSampleShading()
+ minSampleShading()
+ sampleMask()
+ sampleMask()
+ alphaToCoverageEnable()
+ alphaToCoverageEnable()
+ alphaToOneEnable()
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+ alphaFactors()
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+ opaque()
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FCT::VertexLayout
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+ addElement()
+ getElementCount()
+ getElement()
+ getStride()
+ getElementIndexBySemantic()
+ getElementIndexByType()
+ getElementByType()
+ getElementBySemantic()
+ getElementOffset()
+ getElementOffsetByType()
+ getElementOffsetBySemantic()
FCT::OldRenderGraph
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m_needUpdataFramesIndexImages
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+ getResourceImage()
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+ addPass()
+ addWindowResource()
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+ addPassDenpendency()
+ bindOutputImage()
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+ bindDepthStencil()
+ bindOutput()
+ checkAndUpdateResourceSizes()
+ generateImageGraph()
+ computeSize()
+ updateFrameIndices()
+ saveExecutionOrder()
+ generateImageResource()
+ compile()
+ analyzeImplicitDependencies()
+ isPassWritingToResource()
+ isPassReadingFromResource()
+ execute()
+ printExecutionOrder()
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+ getPassByName()
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FCT::ResourceManager
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- m_needChangeIndexImages
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+ allocateImage()
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FCT::RenderGraph
- m_commandBufferIndex
- m_passGroupBarriers
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- m_allocatedPasses
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- m_passGroupOrders
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- m_originPasses
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- m_compiledPasses
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- m_imageToBufferMap
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+ subscribe()
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+ RenderGraph()
+ addPass()
+ compile()
+ executeAllPassGroups()
+ getPass()
+ getImage()
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+ addPass()
+ compile()
+ getImage()
- initForSubmit()
- getOrCreateImageNode()
- getOrCreateImageNode()
- getOrCreateImageNode()
- createTextureEdge()
- createTargetEdge()
- createDepthStencilEdge()
- simulatePassGroupExecution()
- simulateExecutionAndAnalyze Barriers()
- checkBarriersBeforePassGroup()
- executeBarriers()
- addPass()
- resolveTextureSizes()
- groupPasses()
- allocateResources()
- createRHIPasses()
- analyzePassGroupDependencies()
- topologicalSort()
- analyzePassGroupBarriers()
- computePassGroupExecution Order()
- createPassGroups()
- convertShaderStageToPipeline Stage()
- submitPassGroup()
- allocateRootImages()
- allocateImages()
- allocateImages()
- collectImages()
- unionHasSizeBuffers()
- computeBufferSize()
- addNodeToTokenGraph()
- isWaitNode()
- isMutilInputNode()
- isMutilOutputNode()
- partitionPassGroups()
- buildPassGroupGraph()
- createBuffers()
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FCT::Window
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m_x
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m_title
m_needEnableDepthBuffer
m_depthBufferFormat
+ getType()
+ Window()
+ ~Window()
+ pos()
+ size()
+ createPlatform()
+ create()
+ bind()
+ isRunning()
+ swapBuffers()
+ getWidth()
+ getHeight()
+ setCursorPos()
+ registerHandler()
+ unregisterHandler()
+ getCursorPos()
+ clearHandler()
+ targetImage()
+ getCallBack()
+ getCurrentTarget()
+ addRenderFinshSemaphore()
+ clearRenderFinshSemaphores()
+ setPresentFinshSemaphore()
+ addRenderFinshFence()
+ clearRenderFinshFences()
+ initRender()
+ title()
+ enableDepthBuffer()
+ getSwapchainImageCount()
+ getSwapchainSampleCount()
+ swapchain()
+ addModule()
+ getModule()
# getSwapchainTarget()
- addSingleComponent()
- addSingleComponent()
- addSingleComponent()
FCT::AutoViewport
- m_needReviewport
- m_dstViewportWidth
- m_dstViewPortHeight
- m_windowWidth
- m_windowHeight
- m_viewportWidth
- m_viewportHeight
- m_viewportOffsetX
- m_viewportOffsetY
- m_resizeCallBack
- m_passes
+ AutoViewport()
+ AutoViewport()
+ ctx()
+ window()
+ enable()
+ resize()
+ computeViewport()
+ submit()
+ submit()
+ addPass()
+ wndPosToViewportPos()
+ enableForWndAllPass()
FCT::PassResource
m_constBuffers
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+ PassResource()
+ bind()
+ bind()
+ addConstBuffer()
+ addTexture()
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+ create()
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+ markAllDescriptorSetsNeed Recreate()
+ isBound()
FCT::TraditionPipelineState
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+ meshes16
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+ needsUpdate
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+ getType()
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+ addMesh()
+ addMesh()
+ addMesh()
+ addMesh()
+ setPassResource()
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+ submit()
#m_callbackHandler
#m_behavior
#m_delayModuleCreate
#m_swapchain
#m_ctx
#m_vertexShader
#m_pixelLayout
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#m_pixelLayout
#m_binaryCode
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#m_executionOrder
#m_ctx
#m_ctx
#m_currentGraph
#m_defaultGraph
-m_context
#m_resourceManager
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#m_ctx
-m_passUnions
-m_passesUnions
#m_bufferUnions
#m_dependencyGraph
#m_resourceDevice
#m_device
#m_groupGraph
#m_tokenGraph
#m_submitTickers
#m_compiler
#m_generator
#m_imageLoader
#m_modelLoader
#m_syncTickers
#m_resourceDevice
#m_back
#m_front
-m_samplers
-m_elements
-m_textures
-m_elements
#m_autoViewport
#m_wnd
#m_wnd
+vertexShader
+pixelShader
+rasterizationState
+blendState
+depthStencilState
+pixelLayout
+resource
+state
+resourceLayout
+vertexLayout
+ctx